



IP Addressing & Subnetting

Introduction

- You can probably work with **decimal numbers** much easier than with the **binary numbers** needed by the computer.
- Working with binary numbers is time-consuming & error-prone.

Octets

- IP version 4 address is a 32 bit number.
- The 32-bit IP address is broken up into 4 octets, which are arranged into a dotted-decimal notation scheme.
- An octet is a set of 8 bits
- Example of an IP version 4:

172.64.126.52

Thinking in Binary

- The binary system uses only 2 values “0 & 1” to represent numbers in positions representing increasing powers of 2.
- We all are accustomed to thinking & working in the decimal system, which is based on the number 10.

Thinking in Binary *(Cont.)*

- To most humans, the number 124 represents $100 + 20 + 4$.
- To the computer, this number is 1111100, which is $64 (2^6) + 32 (2^5) + 16 (2^4) + 8 (2^3) + 4 (2^2) + 0 + 0$

- Each position in a binary number represents, right to left, a power of two beginning with 2^0 & increasing by one power as it moves left: $2^0, 2^1, 2^2, 2^4$, etc.

Converting to Decimal

- You'll need to convert binary to decimal & vice versa to compute subnets & hosts.
- So, it's time for a quick review lesson in binary-to-decimal conversion.
- There are 8 bits in an octet & each bit can only be a 1 or a 0.

Converting to Decimal *(Cont.)*

- What then do you suppose is the largest decimal number that can be expressed in an octet?

Eight 1's (1111 1111)

Converting to Decimal *(Cont.)*

- What is its equivalent decimal value?

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	1	1	1	1	1
128	64	32	16	8	4	2	1

The binary number 1111 1111 converts into the decimal number:

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

Converting to Decimal *(Cont.)*

- Therefore, the largest decimal number that can be stored in an IP address octet is 255.
- The significance of this should become evident later in this presentation.

IP Address Classes

- IP addresses are divided into 5 classes, each of which is designated with the alphabetic letters A to E.
- Class D addresses are used for multicasting.
- Class E addresses are reserved for testing & some mysterious future use.

IP Address Classes *(Cont.)*

- The 5 IP classes are split up based on the value in the 1st octet:

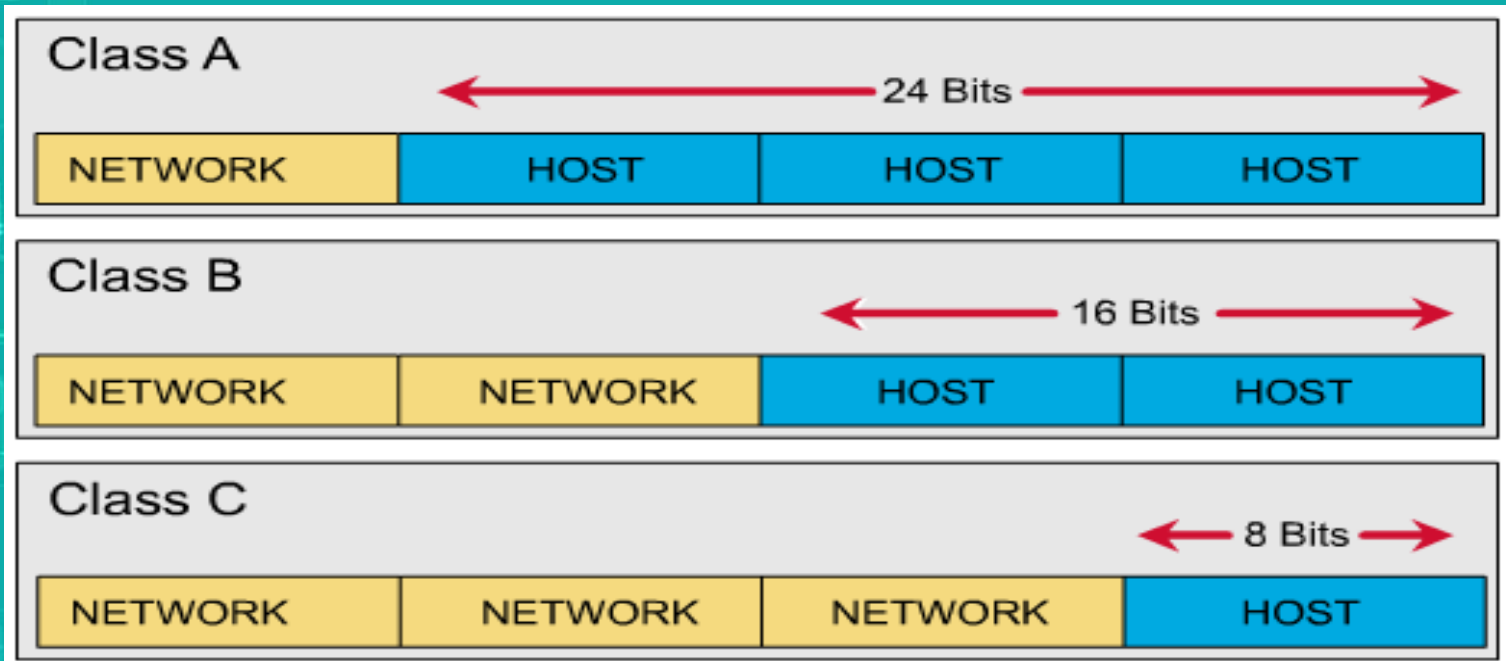
IP Address Class Assignments	
<i>Class</i>	<i>First Octet Value</i>
Class A	0 ~ 127
Class B	128 ~ 191
Class C	192 ~ 223
Class D	224 ~ 239
Class E	240 ~ 255

IP Address Classes *(Cont.)*

- Using the ranges, you can determine the class of an address from its 1st octet value.
- An address beginning with 120 is a Class A address, 155 is a Class B address & 220 is a Class C address.

Are You the Host or the Network?

- The 32 bits of the IP address are divided into Network & Host portions, with the octets assigned as a part of one or the other.



Are You the Host or the Network? *(Cont.)*

- Each Network is assigned a network address & every device or interface (such as a router port) on the network is assigned a host address.
- There are only 2 specific rules that govern the value of the address.

Are You the Host or the Network? *(Cont.)*

- A host address cannot be designated by all zeros or all ones.
- These are special addresses that are reserved for special purposes.

Class A Addresses

- Class A IP addresses use the 1st 8 bits (1st Octet) to designate the Network address.
- The 1st bit which is always a 0, is used to indicate the address as a Class A address & the remaining 7 bits are used to designate the Network.
- The other 3 octets contain the Host address.

Class A Addresses *(Cont.)*

- There are 128 Class A Network Addresses, but because addresses with all zeros aren't used & address 127 is a special purpose address, 126 Class A Networks are available.

Class A Addresses *(Cont.)*

- There are **16,777,214 Host addresses** available in a Class A address.
- Rather than remembering this number exactly, you can use the following formula to compute the number of hosts available in any of the class addresses, where “**n**” represents the number of bits in the host portion:

$$(2^n - 2) = \text{Number of available hosts}$$

Class A Addresses *(Cont.)*

- For a Class A network, there are:
 $2^{24} - 2$ or 16,777,214 hosts.
- Half of all IP addresses are Class A addresses.
- You can use the same formula to determine the number of Networks in an address class.
- Eg., a Class A address uses 7 bits to designate the network, so $(2^7 - 2) = 126$ or there can be 126 Class A Networks.

Class B IP Addresses

- Class B addresses use the 1st 16 bits (two octets) for the Network address.
- The last 2 octets are used for the Host address.
- The 1st 2 bit, which are always 10, designate the address as a Class B address & 14 bits are used to designate the Network. This leaves 16 bits (two octets) to designate the Hosts.

Class B IP Addresses *(Cont.)*

- So how many Class B Networks can there be?
- Using our formula, $(2^{14} - 2)$, there can be 16,382 Class B Networks & each Network can have $(2^{16} - 2)$ Hosts, or 65,534 Hosts.

Class C IP Addresses

- Class C addresses use the 1st 24 bits (three octets) for the Network address & only the last octet for Host addresses. the 1st 3 bits of all class C addresses are set to 110, leaving 21 bits for the Network address, which means there can be **2,097,150** ($2^{21} - 2$) Class C Networks, but only **254** ($2^8 - 2$) Hosts per Network.

Class C IP Addresses (Cont.)

Characteristics of the IP Address Classes						
Class	Address Range	Identify Bits (binary value)	Bits in Network ID	Number of Networks	Bits in Host ID	Number of Hosts/ Network
A	0 ~ 127	1 (0)	7	126	24	16,777,214
B	128~191	2 (10)	14	16,382	16	5,534
C	192~223	3 (110)	21	2,097,150	8	254

Special Addresses

- A few addresses are set aside for specific purposes.
- Network addresses that are all binary zeros, all binary ones & Network addresses beginning with 127 are special Network addresses.

Special Addresses *(Cont.)*

Special IP Addresses			
<i>Network Address</i>	<i>Host Address</i>	<i>Description</i>	<i>Example</i>
0's	0's	Default Cisco Route	0.0.0.0
0's	Host Address	Local Network Hosts	0.0.0.115
1's	1's	Broadcast to Local Network	255.255.255.255
Network Address	1's	Broadcast to Network Address	192.21.12.255
127	Anything	Loopback Testing	127.0.0.1

Special Addresses *(Cont.)*

- Within each address class is a set of addresses that are set aside for use in local networks sitting behind a firewall or NAT (Network Address Translation) device or Networks not connected to the Internet.

Special Addresses *(Cont.)*

- A list of these addresses for each IP address class:

Special Local Network Addresses	
<i>IP Class</i>	<i>Address Range</i>
Class A	10.0.0.0 ~ 10.255.255.255
Class B	172.16.0.0 ~ 172.31.255.255
Class C	192.168.0.0 ~ 192.168.255.255

Subnet Mask

- An IP address has 2 parts:
 - The Network identification.
 - The Host identification.
- Frequently, the Network & Host portions of the address need to be separately extracted.
- In most cases, if you know the address class, it's easy to separate the 2 portions.

Subnet Mask *(Cont.)*

- With the rapid growth of the internet & the ever-increasing demand for new addresses, the standard address class structure has been expanded by borrowing bits from the Host portion to allow for more Networks.
- Under this addressing scheme, called **Subnetting**, separating the Network & Host requires a special process called **Subnet Masking**.

Subnet Mask *(Cont.)*

- The subnet masking process was developed to identify & extract the Network part of the address.
- A subnet mask, which contains a binary bit pattern of ones & zeros, is applied to an address to determine whether the address is on the local Network.
- If it is not, the process of routing it to an outside network begins.

Subnet Mask *(Cont.)*

- The function of a subnet mask is to determine whether an IP address exists on the local network or whether it must be routed outside the local network.
- It is applied to a message's destination address to extract the network address.
- If the extracted network address matches the local network ID, the destination is located on the local network.

Subnet Mask *(Cont.)*

- However, if they don't match, the message must be routed outside the local network.
- The process used to apply the subnet mask involves **Boolean Algebra** to filter out non-matching bits to identify the network address.

Boolean Algebra

- **Boolean Algebra** is a process that applies binary logic to yield binary results.
- Working with subnet masks, you need only 4 basic principles of Boolean Algebra:
 - $1 \text{ and } 1 = 1$
 - $1 \text{ and } 0 = 0$
 - $0 \text{ and } 1 = 0$
 - $0 \text{ and } 0 = 0$

Boolean Algebra *(Cont.)*

- In another words, the only way you can get a result of a **1** is to combine **1 & 1**. Everything else will end up as a **0**.
- The process of combining binary values with Boolean Algebra is called **Anding**.

Default Standard Subnet Masks

- There are default standard subnet masks for Class A, B and C addresses:

Default Subnet Masks	
<i>Address Class</i>	<i>Subnet Mask</i>
Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

A Trial Separation

- Subnet masks apply only to Class A, B or C IP addresses.
- The subnet mask is like a filter that is applied to a message's destination IP address.
- Its objective is to determine if the local network is the destination network.

A Trial Separation *(Cont.)*

- The subnet mask goes like this:
 1. If a destination IP address is **206.175.162.21**, we know that it is a Class C address & that its binary equivalent is:
11001110.10101111.10100010.00010101

A Trial Separation *(Cont.)*

2. We also know that the default standard Class C subnet mask is: **255.255.255.0** and that its binary equivalent is:

11111111.11111111.11111111.00000000

A Trial Separation *(Cont.)*

3. When these two binary numbers (the IP address & the subnet mask) are combined using Boolean Algebra, the Network ID of the destination network is the result:

206.175.162.21 11001110.10101111.10100010.00010101

and

255.255.255.0 11111111.11111111.11111111.00000000

yields

11001110.10101111.10100010.00000000

A Trial Separation *(Cont.)*

4. The result is the IP address of the network which in this case is the same as the local network & means that the message is for a node on the local network.
5. IP address : **206.175.162.21**
6. Subnet mask : 255.255.255.0
7. Network address: **206.175.162.0**

Basics of Subnetting

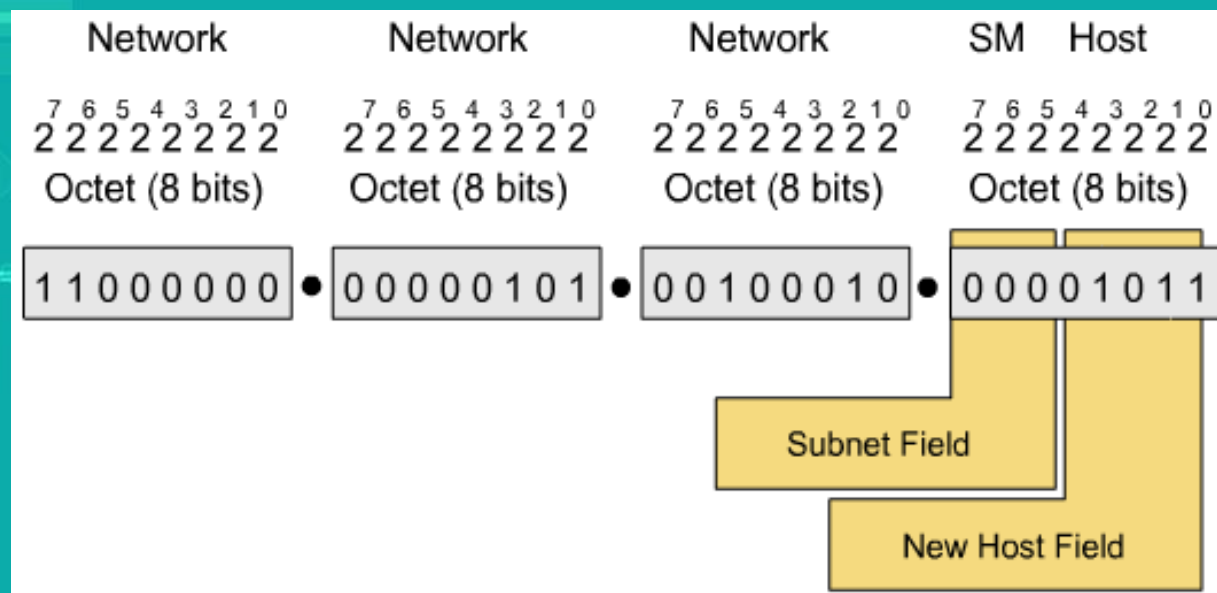
- Classical IP addressing
- Subnetworks
- Subnet mask
- Boolean operations: AND, OR, and NOT
- Performing the AND function

Subnetworks

To create a subnet address, a network administrator borrows bits from the original host portion and designates them as the subnet field.

Subnetworks

SOLUTION: Create another section in the IP address called the subnet.



Subnet Mask

- Determines which part of an IP address is the network field and which part is the host field
- Follow these steps to determine the subnet mask:
 - 1. Express the subnetwork IP address in binary form.
 - 2. Replace the network and subnet portion of the address with all 1s.
 - 3. Replace the host portion of the address with all 0s.
 - 4. Convert the binary expression back to dotted-decimal notation.

Subnet Mask

11111111.11111111.11110000.00000000

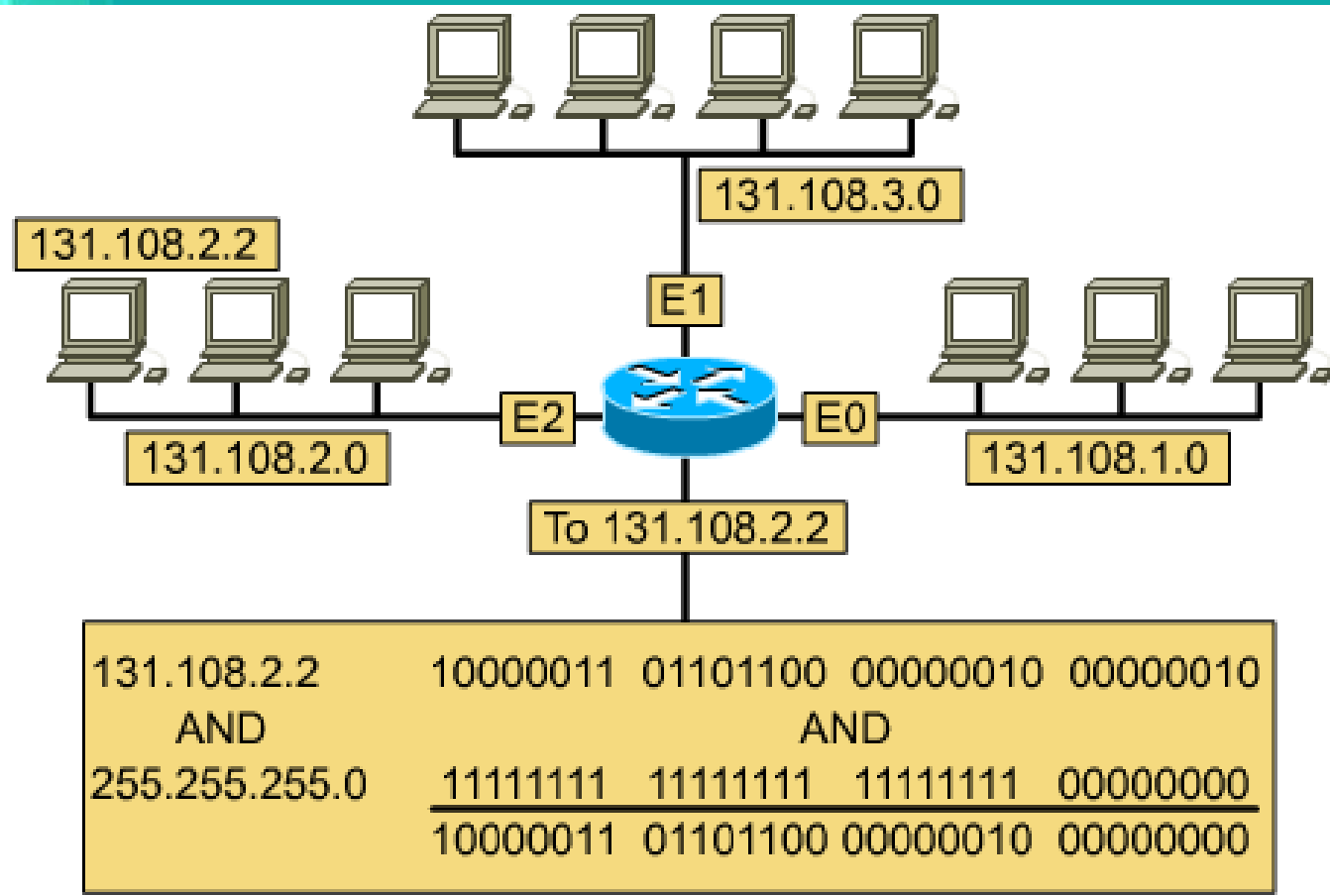
Class B Network
16 bits for the Network
4 bits for the Subnetwork
12 bits for the Host

- ◆ 32 bits long
- ◆ Divided into four octets
- ◆ Network and subnet portions all 1's
- ◆ Host portion all 0's

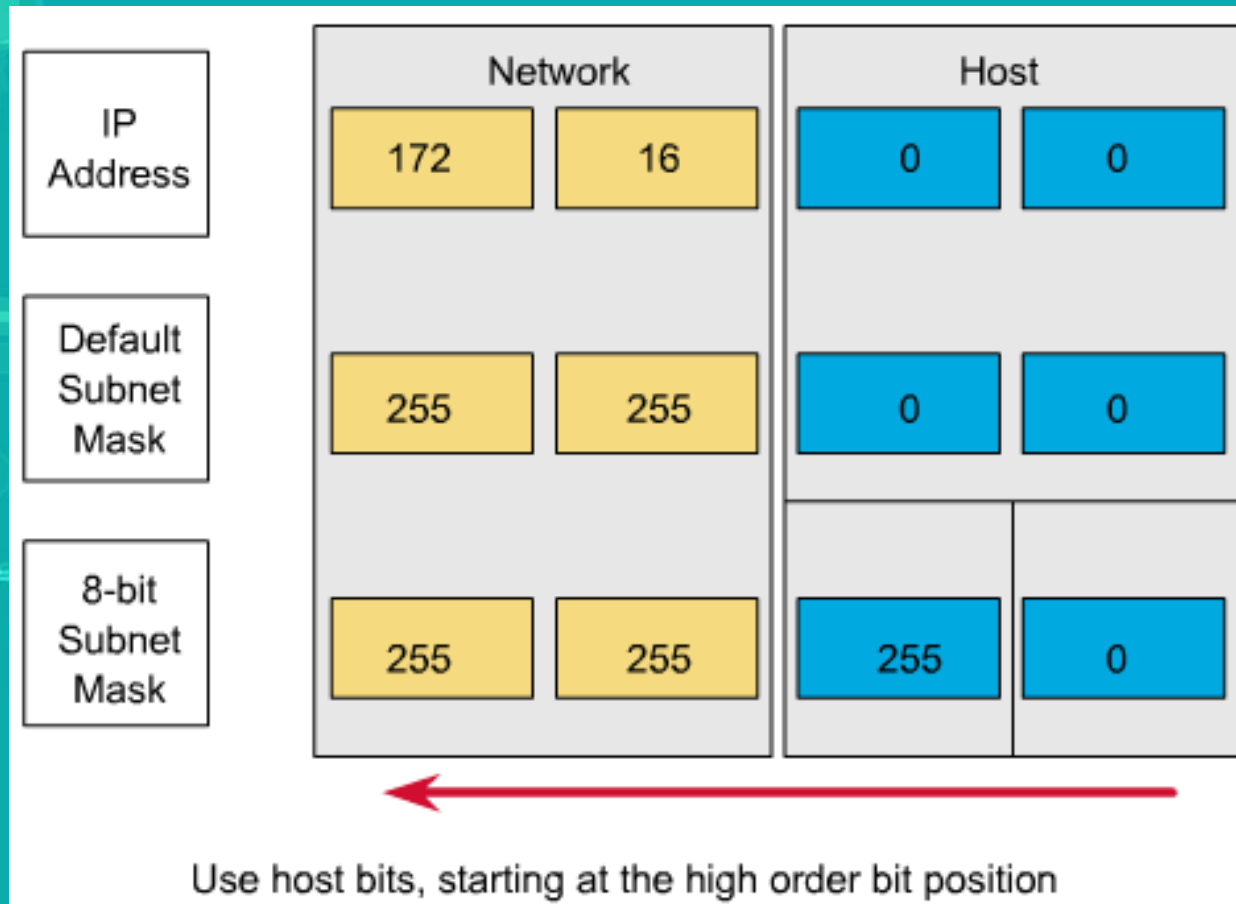
Boolean Operations: AND, OR, and NOT

- AND is like multiplication.
- OR is like addition.
- NOT changes 1 to 0, and 0 to 1.

Performing the AND Function



Range of Bits Needed to Create Subnets



Subnet Addresses

Class B

NETWORK

HOST

Before Subnetting

NETWORK

SUBNET

HOST

After Subnetting

Decimal Equivalents of 8-Bit Patterns

[illegible]

Creating a Subnet

- Determining subnet mask size
- Computing subnet mask and IP address
- Computing hosts per subnetwork
- Boolean AND operation
- IP configuration on a network diagram
- Host and subnet schemes
- Private addresses

Determining Subnet Mask Size

	Network	Subnet	Host
130.5.0.0	10000010 00000101	00000000	00000000
255.255.255.0	11111111 11111111	11111111	00000000
Extended Network Prefix (Subnet Mask)			

Class B address with 8 bits borrowed for the subnet
130.5.2.144 (8 bits borrowed for subnetting) routes to subnet
130.5.2.0 rather than just to network 130.5.0.0.

Determining Subnet Mask Size

Class C address 197.15.22.131 with a subnet mask of 255.255.255.224 (3 bits borrowed)

11000101	00001111	00010110	100	00011
Network Field			SN	Host Field

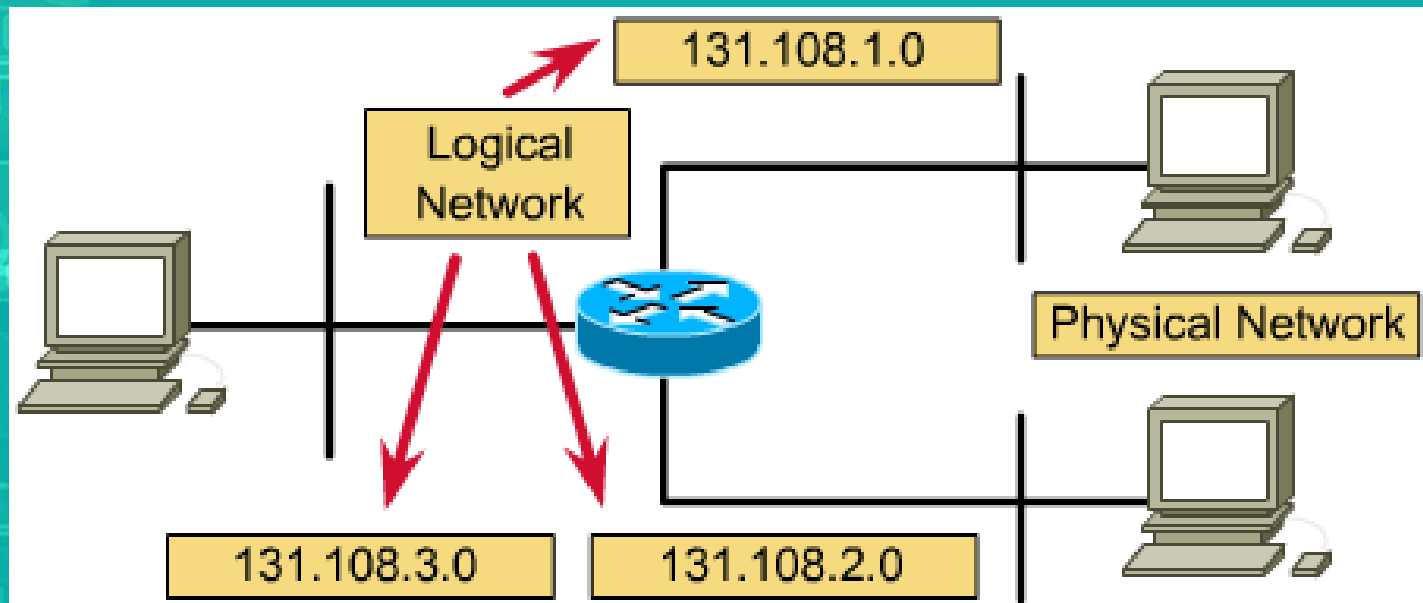
The address 197.15.22.131 would be on the subnet 197.15.22.128.

Subnetting Example with AND Operation

	Network	Subnet	Host
IP Host Address 172.16.2.120	10101100 00010000	00000010	01111000
Subnet Mask 255.255.255.0 or /24	11111111 11111111	11111111	00000000
Subnet	10101100 00010000 172 16	00000010 2	00000000 0

- ◆ Subnet Address = 172.16.2.0
- ◆ Host Address Range = 172.16.2.1 - 172.16.2.254
- ◆ Broadcast Address = 172.16.2.255
- ◆ Eight bits of subnetting

IP Configuration on a Network Diagram



The router connects subnetworks and networks.

Host Subnet Schemes

Number of Bits Borrowed	Number of Subnets Created	Number of Hosts Per Subnet	Total Number of Hosts	Percent Used
2	2	62	124	49%
3	6	30	180	71%
4	14	14	196	77%
5	30	6	180	71%
6	62	2	124	49%

The number of lost IP addresses with a Class C network depends on the number of bits borrowed for subnetting.

Verifying an IP Address

- IP addresses are verified using PING, Trace & Telnet.
- It is important that you know that PING is used to verify IP address connections to the **Network Layer** & that Telnet is used to verify network IP address connections to the **Application Layer**.

Verifying with Telnet

- The reason you need to verify IP addresses is to ensure that the various parts of a network can properly communicate with the other parts.
- Eg., if you can Telnet (**Terminal Emulation Protocol**) into a router from a remote location on the same network, you can verify that the interface & route are up and available.

Verifying with Telnet *(Cont.)*

- Because Telnet operates on the OSI Model's Application Layer, when it's functioning, it's safe to assume that all lower layers are also functioning.

Verifying with PING

- The PING (**P**acket **I**nternet **G**roper) command verifies OSI Layer 3 (Network Layer) connectivity.
- It sends out ICMP (Internet Control Message Protocol) messages to verify both the logical addresses & the Physical connection.

Verifying with PING (Cont.)

- The PING command issued from a Cisco router responds with a number of single character responses.

Cisco PING Response Codes	
<i>Response</i>	<i>Meaning</i>
! (exclamation Mark)	Success
. (period)	Timed out waiting for reply
U	Destination unreachable
 (vertical bar)	Ping process interrupted
? (question mark)	Unknown packet type
C	Congestion-experienced
& (ampersand)	Time to live exceeded

Verifying with Traceroute

- The Traceroute or Trace command is used to show the complete route from a source to a destination.
- Trace sends out probe packets one at a time to each router or switch in the path between the source & the destination IP address entered.

Verifying with Traceroute *(Cont.)*

- Traceroute displays the round-trip time for each packet sent to each upstream router.
- Traceroute has really only 2 results:
 - Time exceeded or
 - Destination unreachable.
- Trace is used to determine where a breakdown in a route may be occurring.

Verifying with Traceroute *(Cont.)*

- Example on how Trace is used:
 - A network has 4 routers (**A**, **B**, **C** & **D**). A Trace command is issued on router **A** to trace the route from itself to router **D**.
 - A timing response comes back from router **B**, but the next message indicates that router **C** is unreachable. You can be fairly certain that the problem lies somewhere on the route between router **B** & router **C**.

Verifying with Traceroute *(Cont.)*

- Like PING, Trace has its own set of response codes:

Trace Command Response Codes	
<i>Response</i>	<i>Meaning</i>
*	Timed out
!H	Router received packet but did not forward it (usually due to an access list)
N	Network unreachable
P	Protocol unreachable
U	Port unreachable

A Short Broadcast

- A broadcast is a message that every node on a network or subnetwork receives & examines.
- Cisco IOS supports 2 different types of broadcast messages:
 - Flooded
 - Directed

A Short Broadcast *(Cont.)*

- Generally speaking, routers do not propagate broadcasts, which is one of the benefits of installing a router in the first place.

A Short Broadcast *(Cont.)*

- Flooded broadcasts (those with the nominal broadcast address of 255.255.255.255) are not forwarded by the router & are considered local traffic only.
- Directed broadcasts, which contain all 1's in the Host portion of the IP address, are addressed to a specific subnetwork & are allowed to pass.

Classless Interdomain Routing (CIDR)

CIDR Background

- Created in response to the exhaustion of IPV4 network addresses
- Increase in size of the Internet's routing tables

Features of CIDR

- Elimination of classful addressing
- Enhanced router aggregation
- Supernetting
- Classless Addressing

Classless Addressing

- Generalised network prefix, could be any length not limited to 8, 16, 24 bits
- E.g. 122.126.66.8/16 identifies a CIDR address with 20 network bits
- Network address is 122.126.0.0
- Broadcast address is 122.126.255.255
- 16 network bits
- 16 host bits

Classless Addressing

- E.g. 172.110.20.2/24
- Network address 172.110.20.0
- Broadcast address 172.110.20.255
- Number of network bits 24
- Number of host bits 8

Variable Length Subnet Masking (VLSM)

- VLSM is a process of dividing an IP network into the subnets of different sizes without wasting IP addresses.
- When we perform Subnetting, all subnets have the same number of hosts, this is known as FLSM (Fixed length subnet mask). In FLSM all subnets use same subnet mask, this lead to inefficiencies.
- Read more:

<http://www.computernetworkingnotes.com/ccna-study-guide/vlsm-tutorial-with-examples.html>

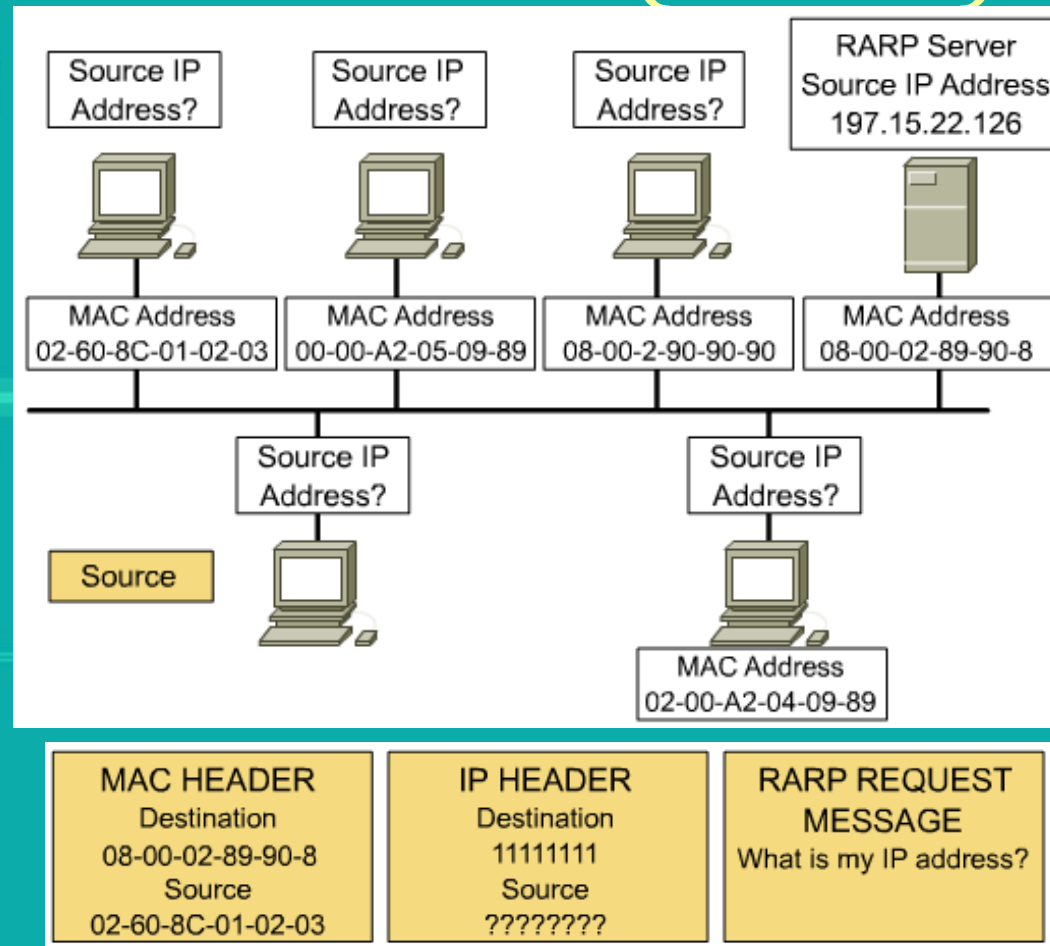
Obtaining an IP Address

- Static addressing
 - Each individual device must be configured with an IP address.
- Dynamic addressing
 - Reverse Address Resolution Protocol (RARP)
 - Bootstrap Protocol (BOOTP)
 - Dynamic Host Configuration Protocol (DHCP)
 - DHCP initialization sequence
 - Function of the Address Resolution Protocol
 - ARP operation within a subnet

Static Assignment of IP Addresses

- Each individual device must be configured with an IP address.

Reverse Address Resolution Protocol (RARP)



The source initiates a RARP request, which helps it detect its own IP address.

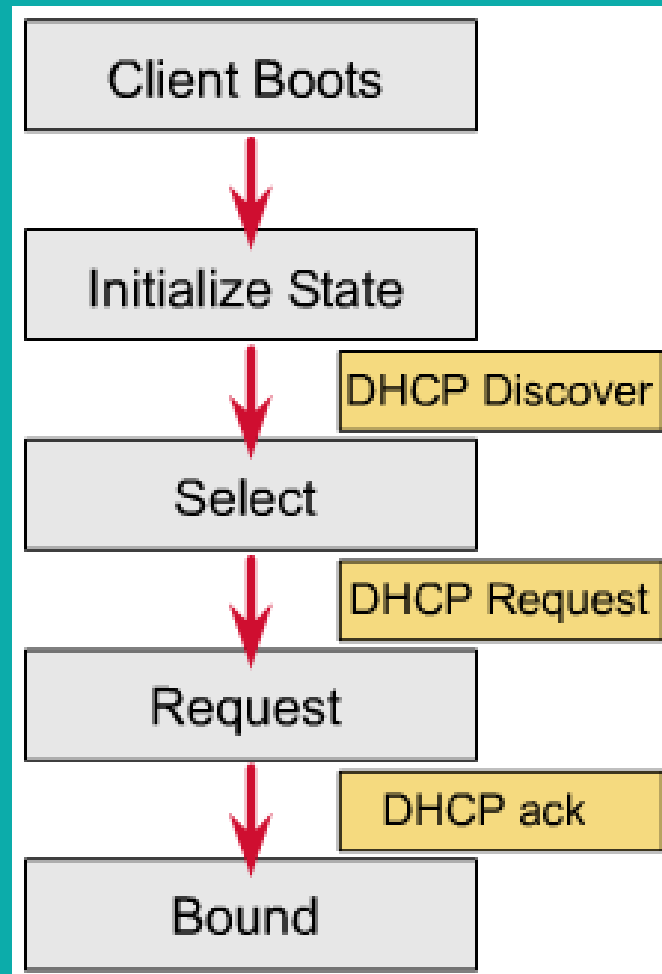
BOOTP IP

- The Bootstrap Protocol (BOOTP) operates in a client/server environment and only requires a single packet exchange to obtain IP information.
- BOOTP packets can include the IP address, as well as the address of a router, the address of a server, and vendor-specific information.

Dynamic Host Configuration Protocol

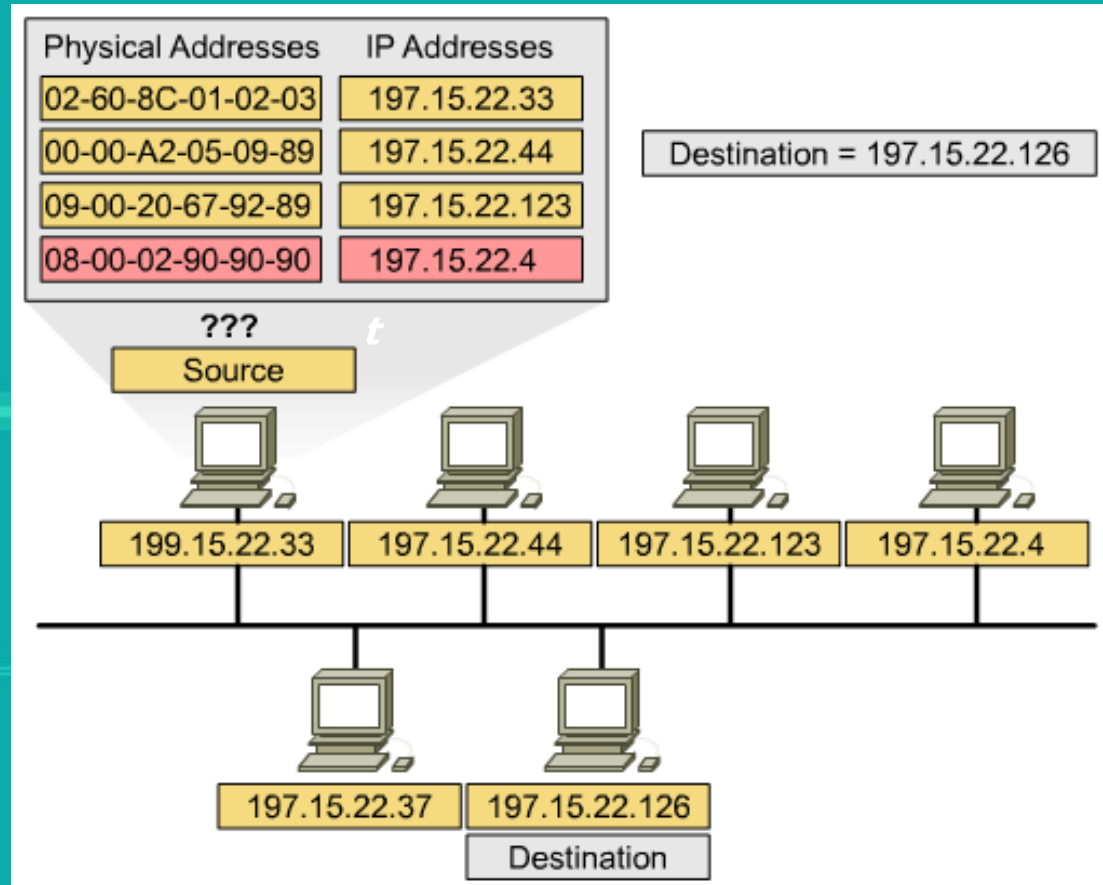
- Allows a host to obtain an IP address using a defined range of IP addresses on a DHCP server.
- As hosts come online, contact the DHCP server, and request an address.

DHCP Initialization Sequence



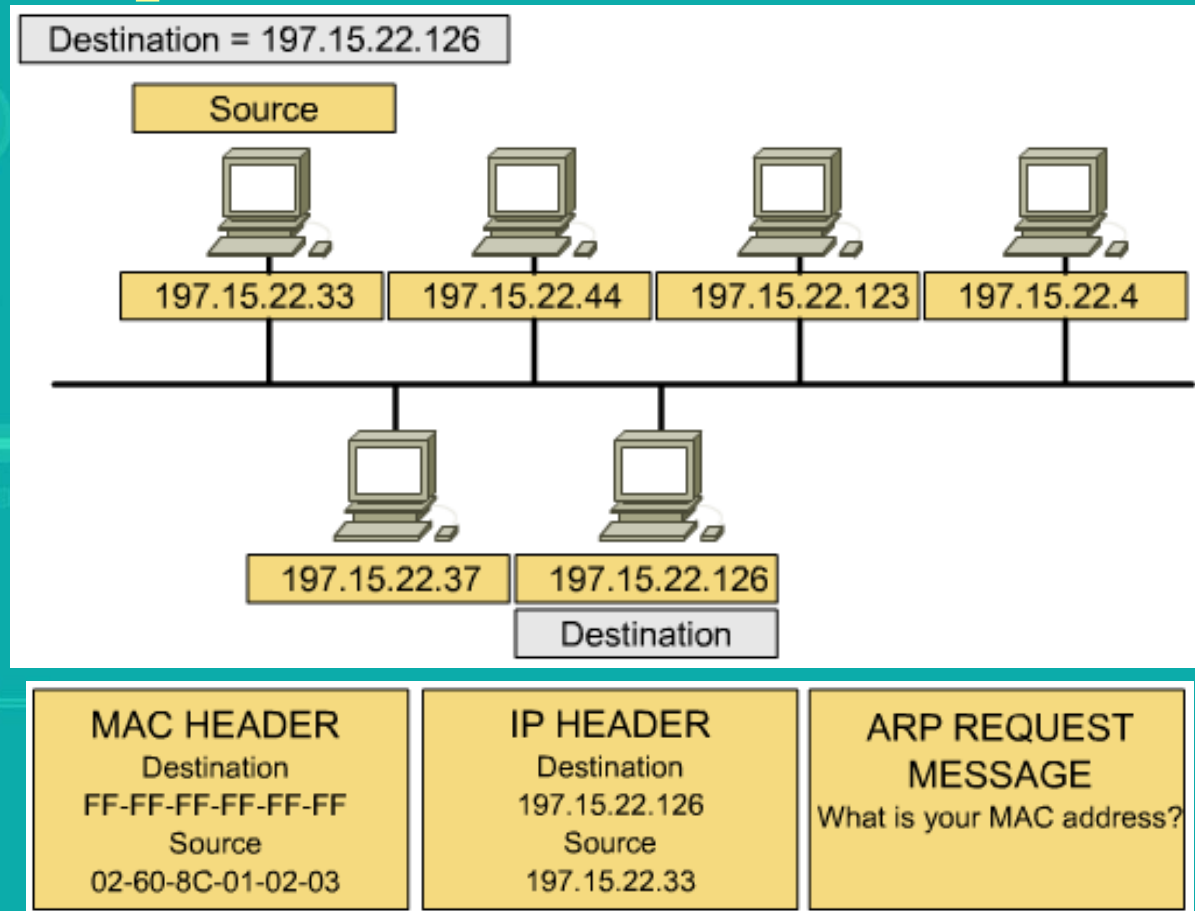
Client collects
DHCP offer
responses from
the server.

ARP



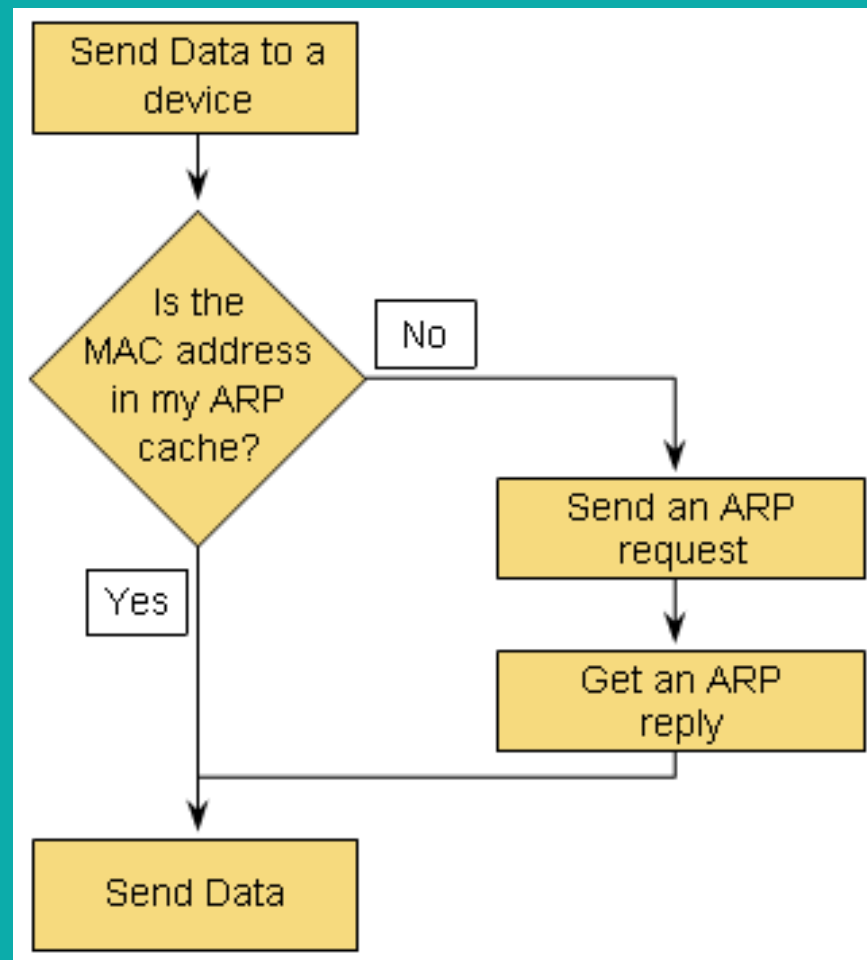
ARP enables a computer to find the MAC address of the computer that is associated with an IP address.

ARP Operation Within a Subnet



All devices on the network receive the packet and pass to network layer; only one device responds with an ARP reply.

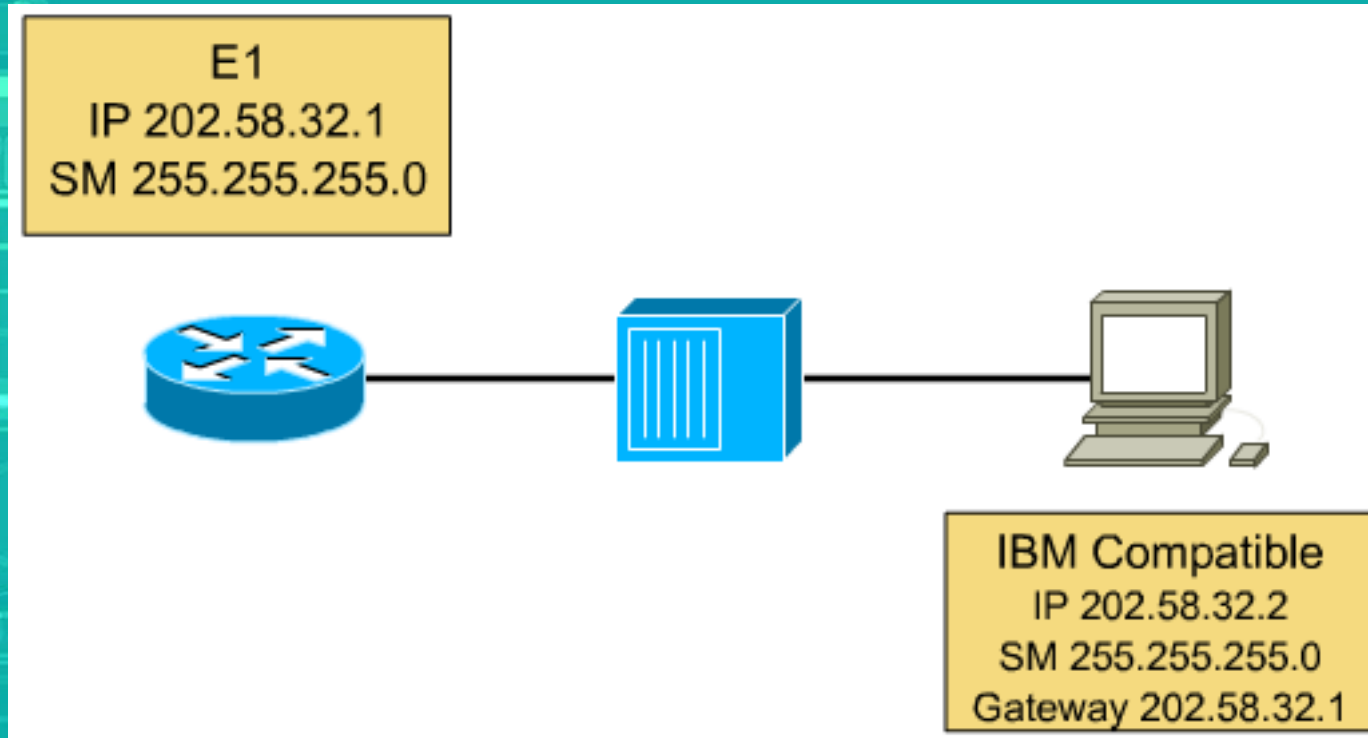
ARP Process



Advanced ARP Concepts

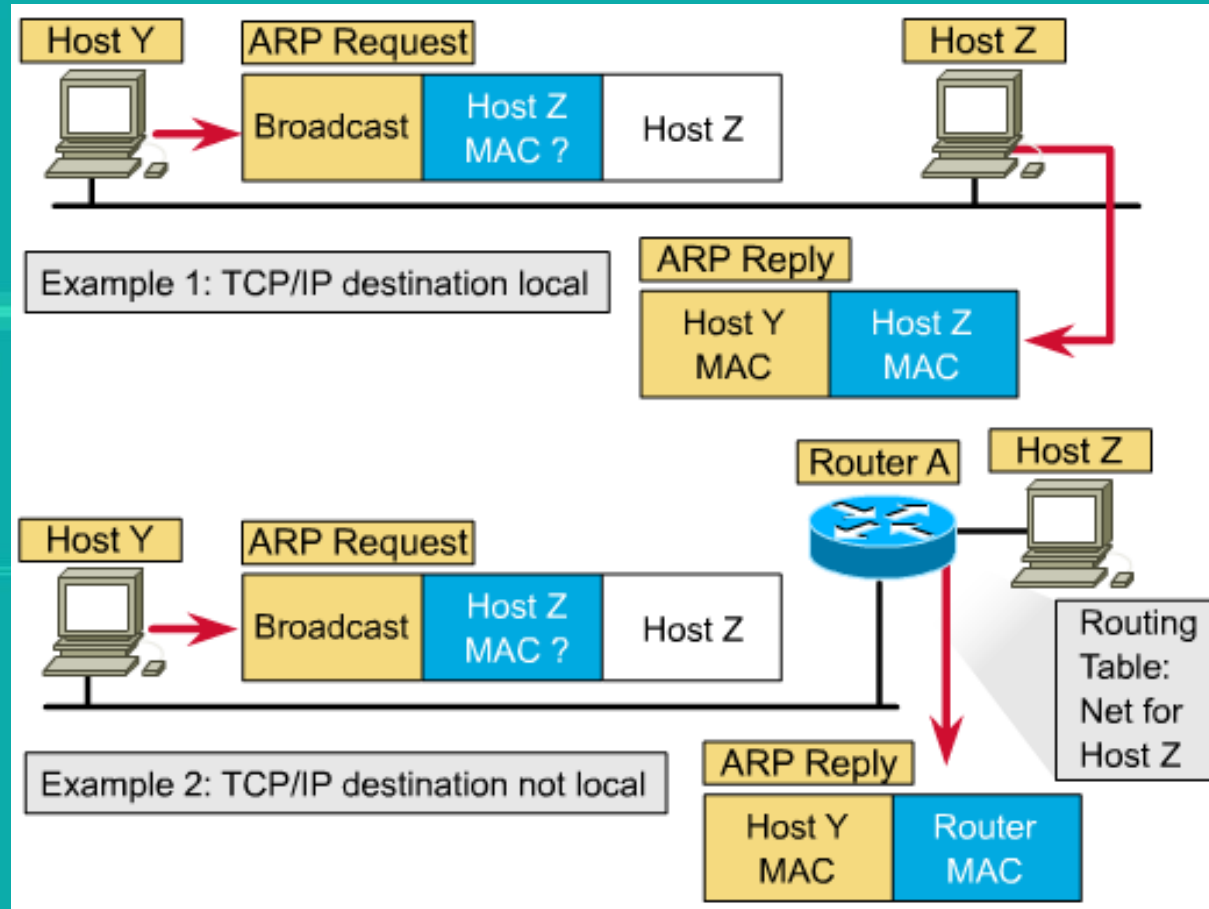
- Default gateway
- Proxy ARP

Default Gateway

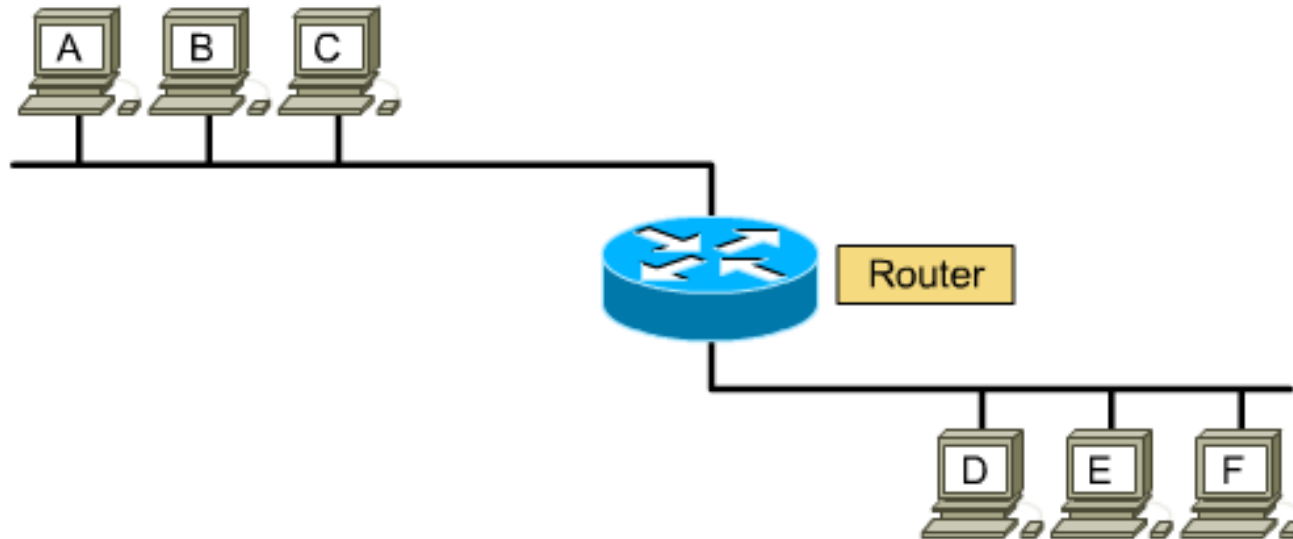


A default gateway is the IP address of the interface on the router that connects to the network segment on which the source host is located.

How ARP Sends Data to Remote Networks



Proxy ARP



- ◆ Machine A sends an ARP request to machine F
- ◆ The router processes and replies to A with its MAC address to be mapped to F's IP address